Predicting PET amyloid in prodromal Alzheimer's disease using Weighted Hidden Markov Model

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Introduction

Quantifying amyloid proteins through PET imaging is the gold standard for assessing Alzheimer's disease (AD). Compared with PET scans, fMRI offers a noninvasive, affordable alternative that can capture dynamic interactions within whole-brain neural networks. Existing work shows relationships between structural MRI and CSF biomarkers [1], and studies have also identified Tau-dependent dynamic Functional Connectivity (dFC) in animal experiments [2]. Compared with deep learning, Hidden Markov Models (HMMs) offer a more interpretable form of dFC. However, modulating continuously changing variables through HMMs is mathematically challenging. In this abstract, our primary goal is to investigate the relationships between PET amyloid concentrations in prodromal AD using HMMs to link PET concentrations with dFC measures from resting-state fMRI data [3]. We present a Weighted Hidden Markov Model (WHMM) that modulates the amyloid-dependent term using a Gaussian mixture, with all formulas rigorously defined using the Expectation–Maximization (EM) algorithm. The proposed model is applied to the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset. Results show that it can successfully characterize amyloid-related changes, yielding a correlation of 0.25 between the real and predicted amyloid values.

Dataset

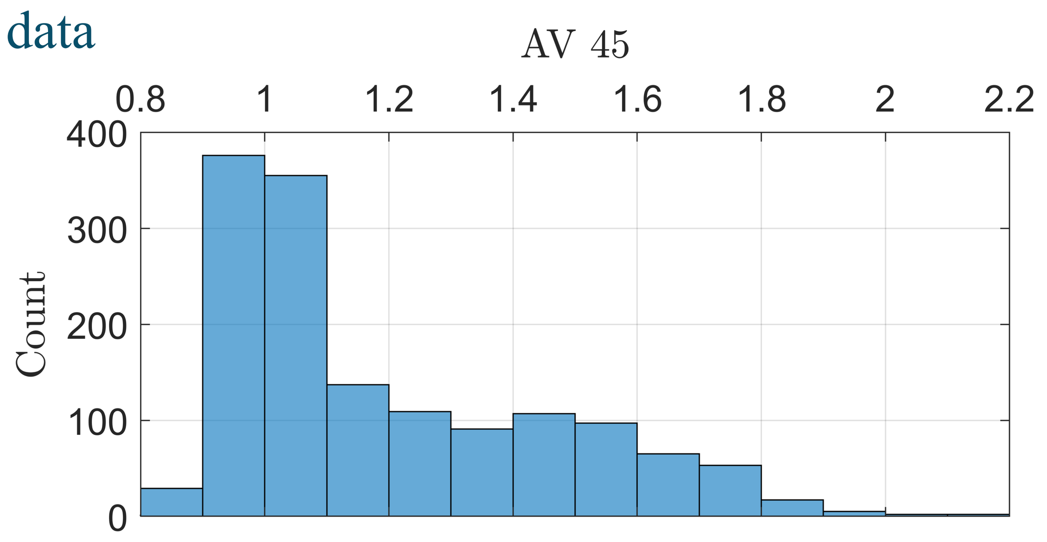
➢ fMRI data[4]

- TR=3s, at least one PET scan performed within 6 months.
- 1446 scans with corresponding AV45 value.

➢ fMRI Preprocessing

(1) Slice-timing correction and rigid-body realignment.
(2) Co-registration.
(3) Spatial normalization to MNI152.
(4) Detrending.
(5) Summation within the CONN 32 mask.
(6) Temporal normalization to zero mean and unit variance.

➢ Amyloid data



➢ Amyloid preprocessing

$$\delta_s = \begin{cases} 0, & \text{AV45} < 0.9 \\ (\text{AV45} - 0.9)/0.9, & 0.9 \le \text{AV45} \le 1.8 \\ 1, & \text{AV45} > 1.8 \end{cases}$$

Weighted HMM

➢ Parameter

Observation: $x_{st} \in \Re^{1\times p}$

Subject index: s

Normalized Amyloid value: δ_s

State occupational probability: $\gamma_{stk} = p(h_t = k|x_{st})$

Emission probability: $p(x_{st}|h_t = k) = (1-\delta_s)P^{(1)}_{skt} + \delta_s P^{(2)}_{skt}$

$$P^{(i)}_{skt} = (2\pi)^{-p/2}\det(\sigma^{(i)}_k)^{-1/2}\times \exp\left[-\frac{1}{2}(x_{st}-\mu^{(i)}_k)^T(\sigma^{(i)}_k)^{-1}(x_{st}-\mu^{(i)}_k)\right]$$

Time length: $T = 135/192/195$

Feature dimension: $p = 32$

Hidden states: $K = 3$

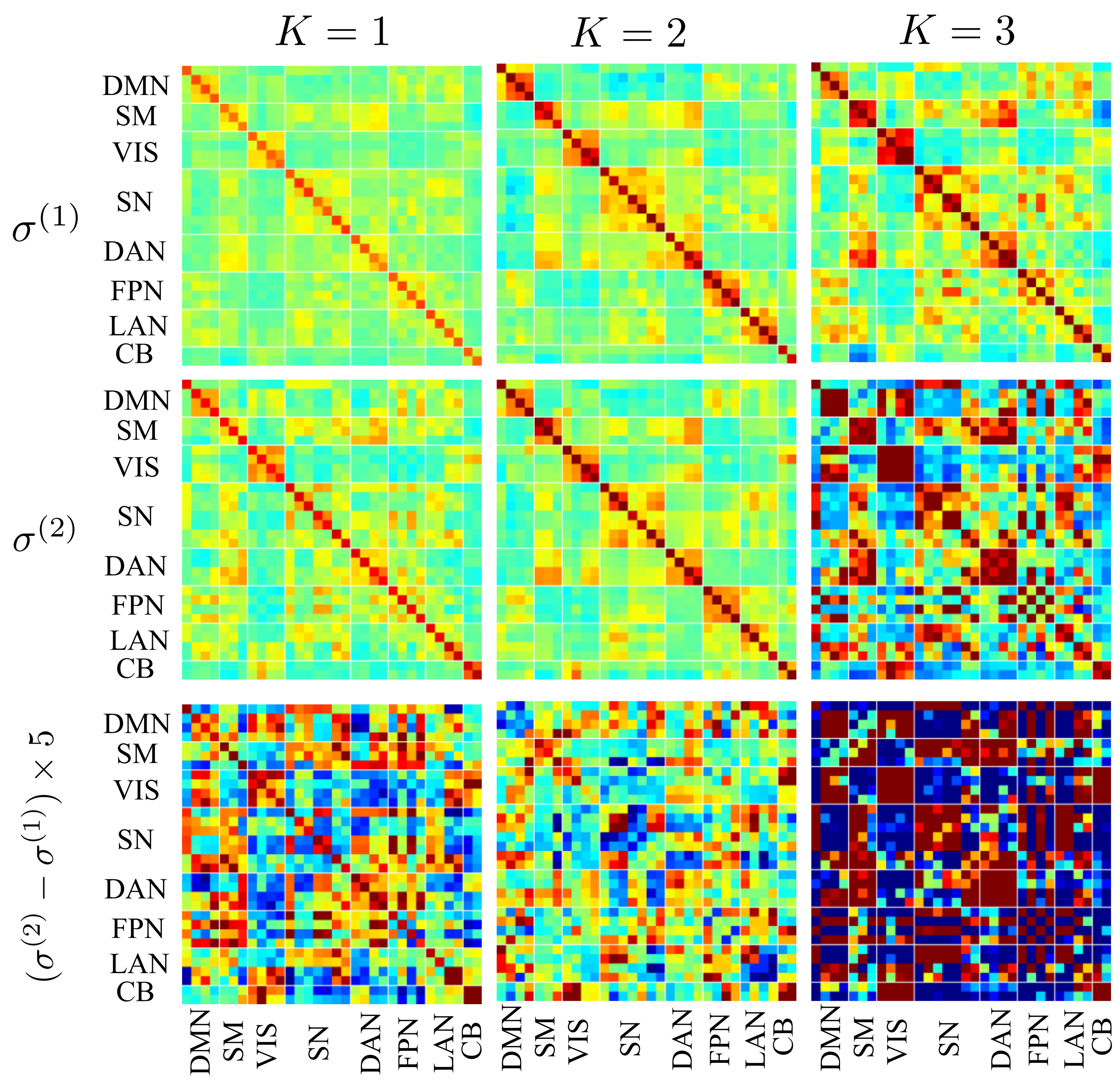
Transition probability: $p(h_t|h_{t-1}) = A_{h_{t-1}h_t}$

➢ EM algorithm [5]

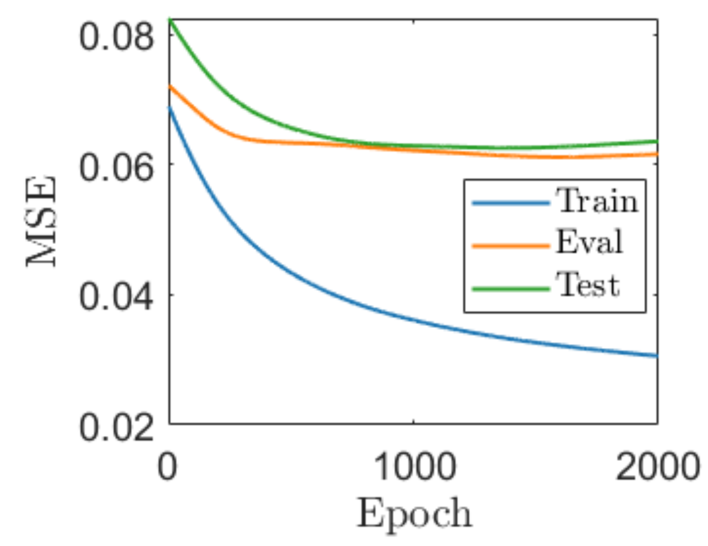
Generative	Training	Initialization: $\{A, \mu^{(1,2)}, \sigma^{(1,2)}\}$	
		Fixed : $\{\delta_{\text{train}} = \delta_{\text{train (real)}}\}$	
		Update (until convergence) : $\{A, \mu^{(1,2)}, \sigma^{(1,2)}\}$	
	Testing	Initialization: $\{\delta_{\text{train}} = \delta_{\text{train (real)}}, \delta_{\text{test}} = 0.5\}$	
		Fixed : $\{A, \mu^{(1,2)}, \sigma^{(1,2)}\}$	
		Update (until convergence) : $\{\delta_{\text{train}}, \delta_{\text{test}}\}$ $\delta_s = \sum_{tk} \frac{\gamma_{stk}}{T}\frac{\delta_s P^{(2)}_{stk}}{(1-\delta_s)P^{(1)}_{stk} + \delta_s P^{(2)}_{stk}}$	
Discriminative	Initialization using generative training results		
	Optimizer: adam	Implementation: PyTorch	
	Learning rate: 1e-4	Loss: $\mathcal{L} = (\delta_{\text{predict}} - \delta_{\text{real}})^2$	

Result

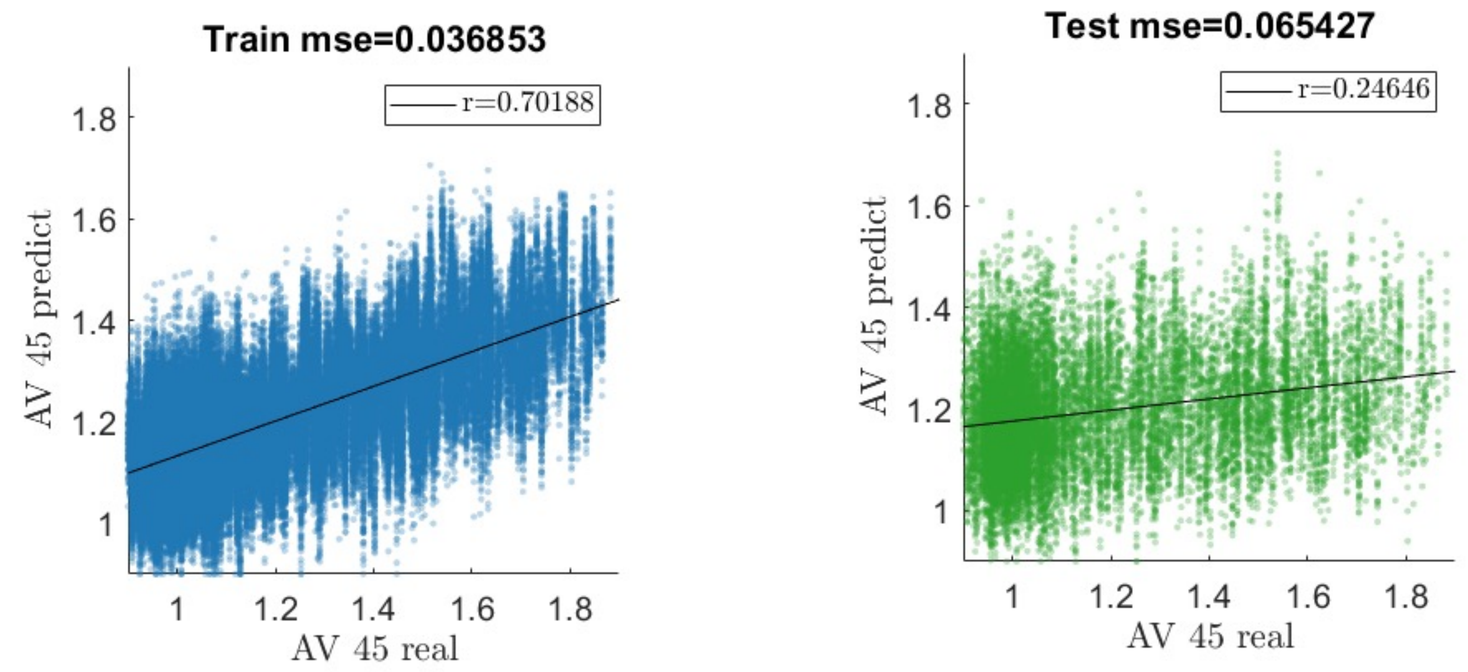
➢ dFC



➢ Discriminative training



➢ Amyloid prediction



Acknowledge

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Reference

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